

Mata Hari 2026

Retro Pin Upgrade to Bally's Mata Hari

Introduction

Retro Pin Upgrade (RPU) is a custom Printed Circuit Board (PCB) and Operating System, by Dick Hamel, that expands the functionality of Solid State pinball machines (Bally, Stern, Williams, Atari, etc.). All the plans and software he has developed for RPU are free & open source (GPL 3.0 license). Detailed descriptions, plans, instructions, and software examples can be found at <https://www.pinballrefresh.com/>

Mata Hari 2026 is a continuation of work originally started by Dick Hamel (MataHari2020) and completed by John Kriz for the Bally Mata Hari machine. The software currently exceeds the size limit for Arduinio Nano and requires the **RPU REV 3** or REV 4 interface board and **Arduinio MEGA 2560 Pro Embed**.

Mata Hari 2026 supports Dual Boot select via the pinball front panel start button and the jumper on the REV 3 or REV 4 interface board. NOTE – Dual Boot support on REV 3 boards is new and only supported in RPU OS version 5.14. For best results, use the entire code set located at <https://github.com/KrizAkoni/MataHari2026Beta>

Self-Test and Audit Settings

Lamps

Credits Display: **00** / Ball in Play Display: **01**

When first entered, all lamps will flash at 2Hz. Pressing the Credit/Reset button will change the Player 1 score to 00 and the first lamp will flash. Repeated pressing of Credit/Reset button will cycle through all lamps (00-59) and then loop back around to 99, which will flash all lamps.

Displays

Credits Display: **XX** / Ball in Play Display: **XX**

When this mode is entered, all displays will cycle through the digits (1-9), changing at 4Hz. Pressing the Credit/Reset button will move through the digits one at a time so they can be tested individually.

Solenoids

Credits Display: **00** / Ball in Play Display: **03**

When this mode is entered, the solenoids (0-14) will be energized one at a time and the solenoid number will be displayed in the Player 1 score box. Each solenoid will be energized for 3/120^{ths} of a second. Pressing the Credit/Reset button will toggle the automatic advance of the solenoid number to energize the same solenoid multiple times in a row.

Switches

Credits Display: **00** / Ball in Play Display: **04**

When this mode is entered, the ID of the first closed switch will be shown in Player 1 score box. Further closed switches will be displayed in Player 2, 3, and 4 boxes. Displays will be blank if there are no closed switches to show.

Sound

FUTURE FEATURE: Credits Display: **00** / Ball in Play Display: **05**

This mode will test the sound features.

Award Score Levels

Credits Display: **00, 01, 02**

The game keeps three Award Score Levels. They are set using the Self Test switch (modes 1, 2, and 3). If an Award Score is set to zero, there is no award given. Otherwise, the award is controlled by the DIP switches (Switch 32) or the Award Score Override (Self Test mode 19). The current Award Score (0, 1, or 2) is shown in the Player 1 score box. During game play, when an Award Score Level is crossed, the player receives the award (extra ball or credit). For tournament scoring mode (when available), the Award Score Levels will be ignored.

In these three setting modes (1, 2, and 3), the scores are changed with the Credit/ Reset button. Pressing the button once increases the score by 1,000 points. Pressing and holding the Credit/Reset button will gradually increase the score. Double-clicking the Credit/Reset button will return the given score to zero. The Award Score Override (Self Test mode 19) can be set to 0-7, or 99. A value of 0-7 overrides the setting of the DIP switches to award either an extra ball or a credit for the given score. A value of 99 turns off the override so the DIP switch (Switch 32) setting will be used.

High Score to Date

Credits Display: **04**

When a player's score exceeds the High Score to Date, the new High Score is recorded and shown during the Attract Mode. Depending on the DIP switch setting (Switch 6), the game will award 3 Credits when a high score is reached. In this mode, the score can be changed with the Credit/Reset button. Pressing the button once increases the score by 1,000 points. Pressing and holding the Credit/Reset button will gradually increase the score. Double-clicking the Credit/Reset button will return the score to zero.

Credits

Credits Display: **05**

This mode allows the operator to see/change the number of credits currently on the machine. The number of credits will appear in the Player 1 score box. Pressing the Credit/Reset button will increase the number of credits up to 20, and then cycle back around to 0.

Audits

Credits Display: **06, 07, 08, 09, 10, 11**

The audit features allow the operator to inspect/reset usage values of the machine. Double-clicking the Credit/Reset button will reset any of the audit values back to zero.

06 - Total plays - total number of games that have been played since this value was reset.

07 - Total replays - total number of credits awarded since this value was reset.

08 - Total times high score was beaten - total number of times the high score has been beaten since this value was reset.

09 - Chute #2 coins

10 - Chute #1 coins

11 - Chute #3 coins

Free Play

Credits Display: **12**

Player 1 score display will show a “0” meaning Free Play is OFF, or a “1” meaning Free Play is ON. The 0 and 1 are toggled by pressing the Credit/Reset button. As soon as the value is toggled, it is written to memory and the machine can be reset into Free Play mode.

If off, the machine will require a positive credit count in order to start a game. Credits can be added through the coin chutes or through setting mode 5. If Free Play is on, hitting the Credit/Reset button will start a game (if in Attract mode), add a player (if ball 1 is in play), or restart the game if ball 2 or more is in play.

Ball Save

Credits Display: **13**

The Ball Save timer is controlled through this mode. The Player 1 score display will read 0, 6, 16, or 21. Pressing the Credit/Reset button cycles through the different values. Setting the Ball Save to “0” turns off the Ball Save feature.

Music Level

Credits Display: **14**

The Credit/Reset button cycles between “0”, “1”, “2”, and “3”. This value controls the amount of sound effects played in the game. Setting to “0” makes the sound effects minimal. A “1” gives some sound, and a “2” gives the full sound. A setting of “3” plays the same level of sounds as “2” but the callouts can be played concurrently. The current level is displayed in the Player 1 score box. If the WAV Trigger add-on board is installed and enabled in the build, the values of 4 and 5 will be added as options. Level 4 uses the WAV Trigger for sound effects and Level 5 adds in background music.

Tournament Scoring

Credits Display: **15**

A value of “0” indicates that Tournament Scoring is off. With a value of “1”, Extra Balls and Credits will not be awarded. Instead, the player will get a point bonus (set by parameters 22 & 23). Pressing the Credit/Reset button toggles between these values.

Reboot

Credits Display: **16**

All displays will show “8007” (meaning BOOT). Pressing the Credit/Reset button will return the machine to Attract Mode.

Skill Shot

Credits Display: **17**

(0 = OFF / 1 = ON) At the start of the ball, the A, B, and SAUCER lights will flash to indicate the skill shot. Hitting the SAUCER when it’s lit will earn 10,000 points.

Tilt Warnings

Credits Display: **18**

The Credit/Reset button cycles between “0”, “1”, and “2”. This parameter sets the number of warnings the player gets before the machine Tilts. At “0”, the machine will tilt at the first hard

nudge. A value of “2” will warn the player twice and tilt on the third hit. A tilt warning consists of a sound.

Award Scores

Credits Display: **19**

The Credit/Reset button cycles through the values 0-7 and then jumps to 99. If set to 99, the DIP switch value is used. Otherwise, the award is based on the table below.

Normally, DIP switch 32 controls what the player gets for achieving an Award Score. In Mata Hari, DIP switch 32 set to “0” awards an extra ball, and “1” awards a credit. This override lets the operator set extra ball or credit for each of the 3 award scores (set in modes 1, 2, and 3) individually.

Award Score Override	Score 1	Score 2	Score 3
0	Extra ball	Extra ball	Extra ball
1	Credit	Extra ball	Extra ball
2	Extra ball	Credit	Extra ball
3	Credit	Credit	Extra ball
4	Extra ball	Extra ball	Credit
5	Credit	Extra ball	Credit
6	Extra ball	Credit	Credit
7	Credit	Credit	Credit
99	(DIP switch)	(DIP switch)	(DIP switch)

Number of Balls Per Game

Credits Display: **20**

Pressing the Credit/Reset button cycles through 3, 5, and 99. A setting of 99 means that the balls per game will be set by DIP switch 16. Otherwise, the balls per game comes from this setting.

Normally, the balls per game (3 or 5) is controlled by DIP switch 16. This setting allows that to be overridden so the operator doesn’t have to go inside the head of the machine to change the value.

Scrolling Scores

Credits Display: **21**

Pressing the Credit/Reset button will toggle this value.

When the player's score exceeds 999,999, the scores will either wrap around to 0 or they will begin to scroll. A setting of "0" in this parameter will make the scores wrap to 0. A setting of "1" will allow the scores to scroll through the 6-digit displays.

Extra Ball Award (for Tournament Scoring)

Credits Display: **22**

The Extra Ball Award is only relevant in Tournament Scoring mode. If Tournament Scoring is activated, collecting an extra ball will give this award instead of Shoot Again. In this mode, the score is changed with the Credit/Reset button. Pressing the button once increases the score by 1,000 points. Pressing and holding the Credit/Reset button will gradually increase the score. Double-clicking the Credit/Reset button will return the given score to zero.

Special Award (for Tournament Scoring)

Credits Display: **23**

The Special Award is only relevant in Tournament Scoring mode. If Tournament Scoring is activated, collecting SPECIAL will give this award instead of a credit. In this mode, the score is changed with the Credit/Reset button. Pressing the button once increases the score by 1,000 points. Pressing and holding the Credit/Reset button will gradually increase the score. Double-clicking the Credit/Reset button will return the given score to zero.

Playfield Valid

Credits Display: **24**

Playfield Valid (0-2) NOT USED

Wizard Mode DURATION

Credits Display: **25**

Seconds (15/30/45/60) - This setting controls the duration (in seconds) of the Wizard Mode. The value can be set to (0, 15, 30, 45 or 60) seconds. A setting of "0" disables Wizard Mode. Pressing the Credit/Reset button will toggle between the values.

Wizard Award

Credits Display: **26**

Score (1,000 and 100,000) - During Wizard Mode, each switch hit will grant the reward dictated by this setting. If set to "0", switches will score no points. The value can be any value between 1,000 and 100,000 points per switch. In this mode, the score is changed with the Credit/Reset button. Pressing the button once increases the score by 1,000 points. Pressing and holding the Credit/Reset button will gradually increase the score. Double-clicking the Credit/Reset button will return the given score to zero.

Dim Level

Credits Display: **27**

Pressing the Credit/Reset button will toggle the dim level between 2 & 3. The Dim Level adjustment allows the operator to choose the duty cycle of "dim" lights. At a level of 2, a dim bulb will be lit 50% of the time. At a level of 3, a dim bulb will be lit 33% of the time. With some bulbs, a noticeable flash will be present at a duty cycle of 33%. With other bulbs, a 50% duty cycle will appear nearly the same as a fully-lit lamp. This adjustment should be re-checked on new machines or if the bulbs are changed. When in this mode, the bonus lamps will toggle between the dim value and the full-lit value (every second). This allows the operator to see the contrast between the two.

RULES

During the course of normal play, the rules of Mata Hari 2026 are similar to the Original game's 1978 rules. New rules and modes have been added as an overlay and are described after this section.

The game will not start until a ball is detected in the trough.

Pressing the start button initiates play. The top of the playfield will flash between A, B, and the SAUCER. Hitting the SAUCER on the 1st shot WHEN LIT by the arrows beneath, scores 1000. Otherwise, the SAUCER scores 3000.

- Rebound switches score 10 points.
- Rolling the A & B orbits (or top lanes) will advance the payout of the A & B orbit (1k, 2k, etc.) The lanes have to be collected symmetrically to advance the reward. Completing A & B 1 time lights one pop-bumper. Making A & B a second time lights both Pop-Bumpers.
- Completing A & B advances the AB Score lamps. AB lit score is collected each time A & B is completed. Pop bumpers are lit with the A & B orbits.
- Pop-Bumpers score 10 points or 100 points when not lit. The outer Pop-Bumpers score 1000 when lit.
- Drop Targets score 500 points. Completing both drop target banks once will score 50,000 points and light the special for the next completion of both banks.
- A base bonus score of 29000 may be accumulated during a ball. A game starts with a bonus of 1000. The bonus score advances one step for each drop target hit, or for the ball passing through the "A" or "B" lanes. Going in the top SAUCER hole gives 3000 bonus each time. Feeder lanes advance 1 bonus step.
- The Bonus Multiplier is advanced by hitting the top SAUCER.
- When the ball enters the outhole the base bonus score is multiplied 2x, 3x, or 5x, and added to the total score. A tilt nullifies the bonus.

NEW MODES

As an overlay to normal play, new modes of game play have been added. These new modes are unlocked by rolling over A and B before the timer in the center of the playfield runs out (10 seconds). If A & B are hit in any order before time runs out, the arrows under the SAUCER will flash, indicating a mode is ready to be started (qualified). Hitting the SAUCER starts the mode. Modes last 30 seconds and are counted down using the BONUS tree. The number of shots left in the goal will flash in the CREDIT display. When a mode times out, it is marked completed if the player scores at least half of the shots in the mode (for Pop Bumpers, all shots are required). If a player completes all the shots, the mode ends regardless of the time remaining. When a mode is finished, the player needs to qualify (A & B within 10 seconds) in order to enter another mode. When all five modes are finished, Wizard Mode begins.

A & B will alternate for orbits mode: Hit A and B as many times as indicated in the blinking CREDIT display (5). As the mode is completed, CREDIT will count down. A mode is considered complete if 50% of the goal is made. Completing the mode awards 5000 points.

A & B reward lights will cycle left for Left Drop Target Mode: Hit the left drop targets as many times as indicated in the blinking CREDIT display (8). As the mode is completed, CREDIT will count down. A mode is considered complete if 50% of the goal is made. Completing the mode awards 5000 points.

A & B reward lights will cycle right for Right Drop Target Mode: Hit the right drop targets as many times as indicated in the blinking CREDIT display (8). As the mode is completed, CREDIT will count down. A mode is considered complete if 50% of the goal is made. Completing the mode awards 5000 points.

Bonus lights will roll up for Pop Bumper Mode: Hit the Pop Bumpers as many times as indicated in the blinking CREDIT display (20). As the mode is completed, CREDIT will count down. Completing the mode awards 5000 points.

Bonus lights will roll down for Slings & In-Lanes Mode: Hit the slings and in-lanes as many times as indicated in the blinking CREDIT display (15). As the mode is completed, CREDIT will count down. A mode is considered complete if 50% of the goal is made. Completing the mode awards 5000 points.

Wizard Mode: Hit as many targets as you can in 30 seconds! A timer and chime will count down for 30 seconds as you rack up as many points as you can. Score values are increased to the Wizard Score setting for every hit (10000 default). Ball Save is also during this mode. At the end of the timer, gameplay is returned to normal and the modes are all reset for another attempt at a round.